What Survives the Ghost Test?
A Four-Level Phase-1 Classification in Pure-Weyl Gravity

GPT-5.6.sol*   Claude Fable 5†

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Abstract

Fourth-order gravity predicts more gravitational-wave solutions than general relativity. Some appear with the wrong sign and are therefore called ghosts, but a wrong sign in an equation is a warning rather than a complete physical verdict. We ask which extra solutions remain after gauge symmetry, global constraints, nonlinear consistency, boundary conditions, and quantum consistency are imposed. Some survive early reductions; different later tests exclude different candidates.

Pure Weyl gravity is a fourth-order conformal metric theory. We separate four logically different objects: local solutions; gauge-, charge-, and boundary-reduced classical directions; nonlinearly continuable directions; and quantum states or particles.

On compact Weyl–Maxwell laboratories, additional axial and polar directions survive local gauge reduction and have nonzero induced Lee–Wald pairings. The signature of the associated positive-frequency Hermitian current form is not organized by the simple rule "Einstein positive, additional negative." At second order, finite-harmonic exponential-polynomial continuation is equivalent to five quadratic stabilizer-charge conditions, whereas bounded continuation obeys further polynomial-growth and characteristic-shell resonance constraints. On Schwarzschild, horizon regularity alone admits additional curvature solutions, while a specified radial Lee–Wald slice-pairing condition excludes the additional axial and polar solutions at the specified fixture $\ell = 2$, $M\omega = 3/5$, where M is the Schwarzschild mass. In the quantum analysis, strict fixed-field-content pure Weyl gravity has a nonzero local Euclidean one-loop BV anomaly. A formal compensator makes the anomaly exact, but changes the theory and does not supply a Lorentzian state space.

We also analyze a changed two-phase counterflow gravity–clock theory. It has a complete free causal BV complex at a specified Berger background and removes a dressed-trace obstruction. Charge reduction reveals an exact tradeoff: making time translation null removes the relative clock, while retaining the clock makes time translation charged. More decisively, the complete $j = \frac{1}{2}$ reduced block contains a Hamiltonian–Hopf quartet throughout the connected trace-healthy same-field stationary family. Phase 1 therefore ends with a classification rather than a selected theory: pole-level signs are too coarse, but the more complete reductions do not generically rescue the candidates tested here. No positive full-BV state, scattering theory, or unitarity theorem is claimed.

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*OpenAI model and principal programme author.

†Anthropic model and computational coauthor. The project was commissioned and coordinated by Asger Alstrup Palm (`asger@area9.dk`), who initiated the questions, exercised editorial control, and serves as corresponding human contact.

1 The question and the four levels

The metric action of strict pure Weyl gravity is

$$S_{\mathrm{W}}[g] = \alpha_{\mathrm{W}} \int_M \sqrt{-g}\, C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}\, \mathrm{d}^4 x, \qquad B_{\mu\nu} = 0, \tag{1}$$

where $B_{\mu\nu}$ is the Bach tensor. Around a fixed background, the linearized Bach equation is fourth order. A formal factorization, a partial fraction decomposition, or a negative residue can reveal an indefinite direction. None of these alone says whether the direction survives gauge reduction, global constraints, physical boundary conditions, nonlinear integrability, or the construction of a quantum state.

This paper organizes the first phase of a computer-assisted research programme by the following hierarchy:

$$\boxed{\begin{array}{ll} \text{Level 1:} & \text{local solutions of the linear field equations,} \\ \text{Level 2:} & \text{reduced classical directions after gauge, charge, and boundary conditions,} \\ \text{Level 3:} & \text{directions tangent to nonlinear solutions in a stated correction class,} \\ \text{Level 4:} & \text{quantum states, particles, and a positive physical probability interpretation.} \end{array}} \tag{2}$$

The arrows between these levels are mathematical problems. They are not terminological promotions. A nonzero classical symplectic pairing at Level 2 is not a positive quantum norm at Level 4. A secular second-order primitive at Level 3 is not nonlinear stability. A local Euclidean anomaly calculation is not a Lorentzian scattering theorem.

The phase-one question is therefore not simply "does the propagator contain a ghost?" It is:

Which additional fourth-order directions survive each declared reduction, which fail nonlinear or boundary tests, and what remains undecided before a particle interpretation can even be posed?

The answer is mixed. Some additional directions survive local gauge and presymplectic reduction. Compact nonlinear constraints remove some of them, while balanced combinations can continue formally with secular terms. A specified Schwarzschild radial-pairing condition removes them at one two-parity fixture. The strict quantum gauge theory is locally anomalous. Finally, a changed theory that passes a difficult causal construction fails a physical spectral test. Different candidates fail at different levels.

2 Theories, conventions, and scope

We use Lorentzian signature $(-,+,+,+)$ and $[\nabla_\mu, \nabla_\nu]v^\rho = R^\rho{}_{\sigma\mu\nu}v^\sigma$. Classical fields are real; complex Fourier modes are bookkeeping devices and physical pairings use the corresponding real or conjugate-frequency form. The connected local gauge group of the strict theory is

$$\mathrm{Diff}_0(M) \ltimes C^\infty(M, \mathbb{R})_{\mathrm{Weyl}}, \qquad g_{\mu\nu} \mapsto e^{2\sigma} g_{\mu\nu}. \tag{3}$$

Unless a theorem card says otherwise, no boundary counterterm or topological density is included in the action. Boundary and support conditions are part of the theorem, not implicit conventions.

Two comparison theories appear. On the compact product we use

$$S_{\mathrm{WM}}[g, A] = S_{\mathrm{W}}[g] - \frac{1}{4}\int_M \sqrt{-g}\, F_{\mu\nu}F^{\mu\nu}\, \mathrm{d}^4x, \tag{4}$$

$$S_{\mathrm{EM}}[g, A] = \frac{1}{2\kappa}\int_M \sqrt{-g}\,(R - 2\Lambda)\, \mathrm{d}^4x - \frac{1}{4}\int_M \sqrt{-g}\, F_{\mu\nu}F^{\mu\nu}\, \mathrm{d}^4x. \tag{5}$$

The magnetic bundle is fixed in that comparison. Einstein–Maxwell is a source theory and comparator; it is not identified with a subsector carrying the same symplectic structure.

The quantum repair uses a formal Wess–Zumino field τ with $s\tau = \omega$, where ω is the Weyl ghost, and the dressed metric $\widehat{g} = e^{-2\tau} g$. This is an enlarged BV theory. It is not strict pure Weyl gravity in another gauge.

The later counterflow experiment changes the classical action as well. Its defining matter mechanism is two positive phase fields with an auxiliary diagonal $U(1)$ connection,

$$S_{\mathrm{phase}} = -\frac{1}{2}\int_M \sqrt{-\widehat{g}}\, \left[f_1^2(\mathrm{d}\theta_1 - A)^2 + f_2^2(\mathrm{d}\theta_2 - A)^2\right]. \tag{6}$$

Writing

$$\chi = \frac{f_1^2\theta_1 + f_2^2\theta_2}{f_1^2 + f_2^2}, \qquad \psi = \theta_1 - \theta_2, \qquad \mu^2 = \frac{f_1^2 f_2^2}{f_1^2 + f_2^2}, \tag{7}$$

separates a diagonal gauge phase from a gauge-invariant relative phase with positive bare kinetic coefficient μ^2. Fixing the harmless positive diagonal normalization to $f_1^2 + f_2^2 = 4$, the complete counterflow action used below is

$$S_{\mathrm{cf}} = \int_M \sqrt{-\widehat{g}} \left[\frac{\alpha_B}{8}\widehat{C}_{\mu\nu\rho\sigma}\widehat{C}^{\mu\nu\rho\sigma} + \frac{M_2}{2}\widehat{R} - V_0 - \frac{\mu^2}{2}(\widehat{\nabla}\psi)^2 - 2(A - \mathrm{d}\chi)^2\right] \mathrm{d}^4x. \tag{8}$$

There is no Maxwell kinetic term for A. Besides diffeomorphisms, the local gauge transformations are

$$g \mapsto e^{2\sigma} g, \quad \tau \mapsto \tau + \sigma, \qquad \chi \mapsto \chi + \lambda, \quad A \mapsto A + \mathrm{d}\lambda, \tag{9}$$

so that $\widehat{g}$ and ψ are gauge invariant. Constant shifts of ψ generate the physical relative symmetry R_{rel} rather than a local gauge symmetry. Equation (8), its BV cotangent lift, and ordinary nonminimal gauge-fixing doublets define the changed theory; a component Hessian is evidence for this action, not its definition. Results for the changed theory are never imported back into the strict theory.

2.1 The scope tuple

The mathematical scope of every result is typed by

$$\boxed{\mathfrak{S} = (\mathcal{T}, M, \bar{\Phi}, \mathcal{G}, \mathcal{X}, \mathcal{B}),} \tag{10}$$

where the entries are the theory, spacetime, background fields, gauge group, function or phase space, and boundary/support class. Evidence is recorded separately as

$$\mathfrak{E} = (\text{claim identifier}, \text{dependency tags}, \text{evidence state}, \text{limitation}). \tag{11}$$

Thus an epistemic label is never treated as part of the mathematical object. Two statements with different entries in $\mathfrak{S}$ are not combined without a constructed map. This matters because the programme uses several controlled laboratories:

Laboratory	Question isolated there
$\mathbb{R} \times S^3$ conformal cylinder	Complete free causal complex and a selected residual zero-charge reduction
Biaxial Berger spacetime	Clocks, relational observables, interactions, and the counterflow successor
$\mathbb{R} \times S^1 \times S^2$ magnetic product	Einstein/additional branches, Lee–Wald pairing, Taub constraints, and resonances
Nariai	Transfer of the causal construction to a second curved background
Schwarzschild and static spherical metrics	Horizons, finite-flux admissibility, and black-hole perturbations
Euclidean S^4 and regular local charts	One-loop determinants and local BV anomaly classes

These are related tests of a common architecture, not successive approximations to one already unified universe.

3 What "ghost" can mean

The literature uses the word *ghost* for inequivalent failures. The distinctions are especially important in a gauge theory with a fourth-order equation. A modern review of the nondegenerate higher-derivative Ostrogradsky theorem is Ref. [4]; gauge degeneracy and physical reduction must be analyzed separately.

Diagnosis	Mathematical question	Phase-1 disposition
Ostrogradsky direction	Is a nondegenerate higher-derivative Hamiltonian unbounded before constraints?	Crosswalks are computed on selected blocks; this is not used as a complete physical quotient.
Negative pole residue	Does a gauge-fixed propagator decompose with a negative coefficient?	Treated as an initial warning, not a final state classification.
Negative classical energy or flux	What is the sign of a declared canonical-energy form, flux, or positive-frequency Hermitian current derived from the real Lee–Wald form?	Computed for selected compact and black-hole sectors. Some signs contradict a naive Einstein/additional split.
Radical or pure gauge	Does the induced presymplectic form vanish on the direction?	Additional compact-product directions are nonradical on the declared pre-residual quotient.
Nonlinear obstruction	Is a linear direction tangent to a second-order solution in a stated function class?	Classified completely on a finite harmonic exponential-polynomial space; bounded continuation is stricter.
Negative quantum norm	Is there a BRST-compatible positive state on the full physical algebra?	Not established. Reduced indefinite two-point functions are not a full-BV state.
Failure of unitarity	Does an asymptotic state space satisfy the optical theorem?	Not tested: no complete scattering or particle theory exists.
Alternative pole prescription	Is the theory redefined by a PT-symmetric, Dirac–Pauli, or fakeon prescription?	Not adopted. These are neighboring programmes with different physical rules, not consequences of our classical reductions.

PT-symmetric treatments of the Pais–Uhlenbeck system, Dirac–Pauli formulations of quadratic gravity, and fakeon prescriptions all demonstrate that "negative residue implies negative-probability asymptotic particle" is not the only proposed logical route [8, 9, 10]. They also demonstrate why the chosen state space and prescription must be declared. Phase 1 follows the ordinary Lorentzian BV/Lee–Wald route and leaves those alternatives as comparisons. The perturbative renormalizability and extra massive-spin-two spectrum that made this question central to higher-derivative gravity go back to Stelle's foundational analyses[5, 6].

4 Which additional linear directions survive reduction?

The compact magnetic product is the cleanest place to compare the Einstein–Maxwell and Weyl–Maxwell linear solution spaces. The computation uses both parities, generic harmonics, exceptional modes, and global blocks.

The sign-bearing object must be distinguished from the underlying presymplectic form. Let Ω_{WM} be the real, antisymmetric Lee–Wald form obtained from the Weyl–Maxwell action [1, 2]. Rank and radical are defined on the real reduced solution space; Ω_{WM} itself has no inertia. After complexifying, fix a common stationary shell

$$e^{-i\omega t}, \qquad \omega > 0, \qquad k = \frac{2\pi n}{L}, \qquad \lambda = \ell(\ell+1), \quad \ell \geq 2, \tag{12}$$

and impose the conjugation reality condition between positive- and negative-frequency coefficients. Here L is the circumference of S^1 and $N_{\ell m} = \int_{S^2} |Y_{\ell m}|^2 \mathrm{d}\Omega > 0$ is the chosen spherical-harmonic normalization. On the positive-frequency space define

$$h_+(u, v) = \frac{\Omega_{\mathrm{WM}}(\bar{u}, v)}{-i\omega L N_{\ell m}}, \qquad L N_{\ell m} > 0. \tag{13}$$

Reality and antisymmetry of Ω_{WM} make h_+ Hermitian. A positive change of the circle or harmonic normalization, and a nonzero complex rescaling of any basis vector, act by Hermitian congruence and leave its inertia invariant. The signatures below refer only to this h_+. Exceptional dipoles, generalized zero modes, and $\omega = 0$ are classified in separate real blocks and are not assigned these frequency-shell signatures.

Result card A: compact Einstein/additional decomposition

Theory. Einstein–Maxwell source and Weyl–Maxwell target.

Background. The magnetically supported compactified Plebanski–Hacyan product $\mathbb{R} \times S^1 \times S^2$, in a fixed magnetic bundle.

Domain. Parity-complete generic linear modes, together with the declared exceptional and global blocks.

Gauge and quotient. Local gauge quotient and maximal pre-residual H^0 exact sequence; final stabilizer reduction is not included.

Boundary condition. Spatial periodicity; no asymptotic scattering condition.

Statement. The Einstein–Maxwell solutions inject into the Weyl–Maxwell solutions. The cofiber contains additional axial and polar blocks. On each generic additional real block Ω_{WM} has zero radical and rank four. On the associated positive-frequency complex two-plane, h_+ has inertia $(2, 0)$; its restriction to the target Einstein image has inertia $(1, 1)$ and the complete generic target has inertia $(3, 1)$. The independently normalized Einstein–Maxwell source Hermitian current form has inertia $(2, 0)$, so inclusion does not define a cyclic or symplectic equivalence.

Not claimed. No positive energy, particle norm, final residual quotient, causal boundary admissibility, or nonlinear closure follows.

The important conclusion is not that the additional block is “healthy” because the displayed frequency form is positive. The conclusion is that it is not an algebraic duplicate or a radical direction on this quotient, and that the standard pairing does not respect the naive expectation that the Einstein image should inherit the ordinary Einstein sign while the complement should carry the bad sign. The action matters: the Einstein–Hilbert and Weyl Lee–Wald forms are different even on the same metric perturbation. Neither the rank statement nor the inertia of h_+ is by itself a positive-energy or positive-norm quantum theorem.

This exact-sequence formulation also prevents an unjustified direct-sum claim. A lift of an additional curvature representative may be shifted by an Einstein solution. The invariant information is the injection, cofiber, induced ranks, radicals, and extension class. A canonical branch split requires more structure than composition of differential operators.

On the conformal cylinder a different reduction occurs. In a selected derived zero-charge sector, all fifteen residual conformal generators are implemented by a BFV/Koszul receiver, and the vacuum and selected residual one-particle cohomologies are acyclic. The first surviving H^4 classes are deformation or vertex classes, not one-particle gravitons. This is a theorem about that selected compact sector. It is not a universal removal of radiative gravitons and it supplies no asymptotic particle interpretation.

The two laboratories therefore teach complementary lessons: additional directions can survive a compact local-gauge quotient, while a stronger zero-charge residual reduction can remove selected one-particle cohomology. Neither statement transfers to the other without a map.

5 Which directions survive nonlinear integrability?

Linearization stability on compact Cauchy surfaces has a classical history from Brill–Deser through Fischer–Marsden, Moncrief, and the Arms–Marsden–Moncrief momentum-map normal form [14, 15, 16, 17]. The general idea that a symmetric solution has a quadratic momentum-map cone is therefore not new. The contribution here is an exhaustive model-specific calculation for a fourth-order Weyl–Maxwell mode space and a separation of formal secular solvability from bounded solvability.

Let L be the linearized equation and $D^2E[u,u]$ the quadratic source. Two correction spaces are kept distinct:

$$\mathscr{X}_{\mathrm{ep}} = \left\{\text{finite sums of } t^p e^{-i\Omega t} \text{ with spatially periodic finite harmonic support}\right\}, \tag{14}$$

$$\mathscr{X}_{\mathrm{qp}} = \left\{\text{finite bounded quasiperiodic sums } e^{-i\Omega t}\right\}. \tag{15}$$

The first allows secular polynomials; the second does not.

Result card B: finite-harmonic second-order cone

Theory. Compact Weyl–Maxwell theory, with the Einstein–Maxwell comparison used to label branches.

Background. $\mathbb{R} \times S^1 \times S^2$ with the fixed magnetic bundle.

Domain. The complete declared finite harmonic space: generic Einstein and additional primaries, both parities, exceptional dipoles, twists, homogeneous generalized modes, electric flux, and Wilson directions.

Obstruction covectors. The five stabilizer covectors generated by time translation H, circle translation P_x, and rotations J_1, J_2, J_3.

Correction class. Finite exponential-polynomial corrections $\mathscr{X}_{\mathrm{ep}}$.

Statement.

$$\mathcal{Z}_2^{\mathrm{ep}} = \{u : \mu_H(u) = \mu_{P_x}(u) = \mu_{J_1}(u) = \mu_{J_2}(u) = \mu_{J_3}(u) = 0\}.$$

The five moment maps are necessary and sufficient in this correction class.

Not claimed. This is not a theorem for all smooth or Sobolev fields, retarded solutions, bounded dynamics, all-order integration, or stability.

The sufficiency statement follows from blockwise constant-coefficient surjectivity: nonzero invariant factors are surjective on finite exponential-polynomial modules, including at characteristic roots after the appropriate polynomial degree is allowed. The only zero invariant factors are the five stabilizer covectors. This is the explicit exhaustive content beyond the general momentum-map normal form.

Bounded continuation has a larger obstruction ledger. If the quadratic source is expanded by frequency shell and polynomial degree, then a bounded primitive exists only when three types of functional vanish:

$$\{\mu_X\} \cup \{\mathcal{P}_{j,r}\} \cup \{\mathcal{R}_{j,a}\}. \tag{16}$$

Here μ_X are the global zero-block Taub constraints, $\mathcal{P}_{j,r}$ are coefficients that would require positive powers of t, and $\mathcal{R}_{j,a}$ are adjoint-kernel pairings on nonzero characteristic shells. Their common nonlinear zero locus is not fully classified for arbitrary finite support. It is classified on several important subspaces. In particular, on the complete global plus ($\ell = 2, k = 0$) additional-primary space, the bounded cone reduces to the standard global generalized-zero cone: no nonzero additional-primary amplitude survives. At other momenta, balanced sheets can remain.

The distinction is physical. A resonant source can have a formal primitive $te^{-i\omega t}$ and hence define a second-order jet, while failing every bounded quasiperiodic test. The programme therefore does not report "second-order solvable" without naming the correction class.

The same calculation also blocks a common branchwise simplification. Balanced Einstein–additional configurations can have zero total moment map while their projected Einstein and additional pieces carry equal and opposite nonzero charges. Consequently the ambient exact sequence does not descend to separately neutral branch fibers. Nonlinear reduction is intrinsically mixed.

6 Which boundary conditions retain the extra solutions?

Compact charge constraints and black-hole endpoint conditions test different questions. The Schwarzschild analysis uses the Ricci perturbation $\psi_{\mu\nu} = \delta R_{\mu\nu}[h]$ as a second-order curvature variable in the fourth-order Bach system. On a Ricci-flat background the universal linear identity is

$$\delta B_{\mu\nu} = \frac{1}{2}\Box\delta R_{\mu\nu} + C_{\mu\rho\nu\sigma}\delta R^{\rho\sigma} - \frac{1}{6}\nabla_\mu\nabla_\nu\delta R - \frac{1}{12}g_{\mu\nu}\Box\delta R. \tag{17}$$

The Einstein image lies in $\ker\delta R$; the additional curvature content is visible in ψ.

Horizon analyticity does not force ψ to vanish. Regular ingoing additional curvature solutions exist in the declared real-frequency axial analysis. This already shows that future-horizon regularity alone does not select the Einstein image. It does not show that every local differential initial or boundary condition fails: imposing ψ and its normal derivative to vanish on a Cauchy surface is a local restriction that selects the Einstein image at linear order.

The outer boundary gives a sharper fixture-level result.

Here the Schwarzschild mass is denoted by M and normalized to $M = 1$, so the horizon is $r = 2M = 2$ and the quoted frequency means $M\omega = 3/5$. In ingoing Eddington–Finkelstein coordinates $(v, r, \vartheta, \varphi)$ let $F^v[u_1, u_2](r)$ denote the sphere integral of the v-component of the action-derived Lee–Wald current for two formal mode representatives. For a conjugate-frequency pair define the truncated radial slice form

$$\mathcal{N}_R[u_1, u_2] = i\int_2^R F^v[\bar{u}_1, u_2](r)\,\mathrm{d}r, \qquad \mathcal{N}_\infty[u_1, u_2] = \lim_{R\to\infty}\mathcal{N}_R[u_1, u_2], \tag{18}$$

whenever the improper integral exists in the declared formal asymptotic class. "Finite radial slice form" below means finiteness of this quadratic form. It is neither an L^2 norm on monochromatic scattering states nor a wave-packet Hilbert norm, and the polar zero-sector quadratic form can vanish.

The reconstruction makes the power test explicit. For both parities the homogeneous Einstein reconstruction is governed by a common master variable $F = C_h = H_1'$ with

$$F_E^{(0)} \sim r^{-3}(1 + O(r^{-1})), \qquad F_E^{(2)} \sim e^{-2i\omega r} r^{-4i\omega+1}(1 + O(r^{-1})). \tag{19}$$

The sourced additional metric lifts are different: the axial lifts have a nonzero single-log coefficient in both characteristic sectors; the polar $\mu = 0$ lift is enhanced by one power and has one logarithm, whereas the polar $\mu = -2\omega$ lift is a pure oscillatory power. At the fixture the resulting sphere-integrated densities are

$$\begin{array}{c|ccc} & E\times E & E\times X & X\times X \\ \hline \text{axial} & r^{-2} & r^0 \text{ or } r^1 & r^0 \log r \text{ or } r^2 \\ \text{polar} & 0 \text{ or } r^{-2} & r^1 \text{ or } r^3 & r^2 \text{ or } r^4. \end{array} \tag{20}$$

Thus the finite/divergent distinction is a radial-integrability statement about the full current, not an inference from a characteristic exponent alone.

Result card C: radial-pairing selection at a Schwarzschild fixture

Theory. Strict pure C^2 gravity.

Background. Schwarzschild exterior with $M = 1$ and horizon $r = 2M = 2$.

Domain. Axial and polar $\ell = 2$ reconstructed metric solutions at real dimensionless frequency $M\omega = 3/5$, with the radial slice form (18).

Gauge and quotient. Regge–Wheeler/Zerilli-type parity reduction and the action-derived pure-Weyl Lee–Wald current.

Boundary condition. Future-horizon ingoing regularity followed by finiteness of $\mathcal{N}_\infty$ in the declared formal asymptotic class.

Statement. At this two-parity fixture the Einstein solutions have finite radial slice form, whereas the reconstructed additional axial and polar solutions give divergent radial integrals. This condition therefore selects the Einstein sector at the stated $(\ell, M\omega)$.

Not claimed. No Cauchy-slice Hilbert norm, wave-packet completion, all-frequency theorem, full exterior initial-boundary well-posedness, quasinormal spectrum, ringdown, stability, particle interpretation, or universal causal-truncation theorem follows.

This result is deliberately narrower than the proposition that "ordinary boundary conditions always remove the additional branch." The latter would require a complete asymptotically flat Bach phase space, metric reconstruction and Jordan analysis at all relevant frequencies, finite flux, and a well-posed exterior problem. Existing AdS/Euclidean Einstein-selection arguments operate in different boundary classes [18, 19]. The present contribution is a Lorentzian Schwarzschild horizon/radial-integrability comparison at an exact declared fixture. Covariant phase-space and canonical-energy methods provide the conceptual comparison, but their global hypotheses are not asserted here [2, 3].

7 Can a positive quantum state be constructed?

There are three separate quantum objects in the programme: local BV anomaly cohomology, Euclidean one-loop coefficients, and reduced Lorentzian two-point functions. They are not interchangeable.

In four dimensions a local Weyl breaking at ghost number one and form degree four has essential type-B and type-A representatives of the form

$$\mathcal{A}^{(1)} = \frac{1}{(4\pi)^2} \int_M \sqrt{g}\, \omega \big(c_{\mathrm{W}} C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} - a_{\mathrm{W}} E_4 + b_{\mathrm{W}} \Box R\big)\, \mathrm{d}^4 x. \tag{21}$$

In the action normalization of Eq. (1) and the repository's $\Box R = 0$ finite-counterterm convention, the computed coordinates are

$$\boxed{c_{\rm W} = \frac{199}{30}, \qquad a_{\rm W} = \frac{87}{20}, \qquad b_{\rm W} = 0.} \tag{22}$$

The $\Box R$ coefficient is scheme dependent and may be shifted by a local counterterm. The nonzero C^2 and Euler representatives are the essential type-B and type-A classes, respectively, in the Deser–Schwimmer classification[30]. A one-loop divergence, beta function, trace anomaly, and BV master-equation breaking are related only after the gauge-fixed determinant, normalization, descent, and quotient maps have been supplied.

The complete local determinant and heat-kernel ledger is:

sector	operator	statistics	Z exponent	c_i	a_i
metric TT, upper	$\Delta_{2,\perp}(4)$	bosonic	$-1/2$	37/24	31/72
Diff–Weyl scalar ghost	$\Delta_0(-4)'$	fermionic	$+1/2$	89/120	269/360
metric TT, lower	$\Delta_{2,\perp}(2)$	bosonic	$-1/2$	29/8	181/72
transverse Diff ghost	$\Delta_{1,\perp}(-3)'$	fermionic	$+1/2$	29/40	79/120
exact sum				199/30	87/20

The listed c_i and a_i are net signed contributions: the statistics sign and the displayed determinant exponent have already been applied. They are therefore not the unsigned heat-kernel coefficients of the individual operators. All four factors are second order. Their bundle ranks are $5, 1, 5, 3$; the primes delete five scalar and ten transverse conformal-Killing ghost zero modes. York/Hodge Jacobians and the nonminimal doublets are already included. The table lists a_i with the sign convention $c_i C^2 - a_i E_4$.

The map from this Euclidean spectral ledger to a gauge anomaly is a residue and cohomology map, not a comparison of names. If $I_i(d)$ are the dimensionally continued integrated carriers, then

$$s_d I_i(d) = (d-4)A_i^4 + s_d B_i(d) + \mathrm{d}C_i(d) + O((d-4)^2). \tag{23}$$

Multiplication by the one-loop pole, replacement of the Weyl parameter by the ghost ω, addition of the diffeomorphism completion, and reduction modulo s and d give

$$\Gamma^{(1)}_{\rm div}(d) \xrightarrow{s_d,\ \mathrm{Res}_{d=4}} \mathcal{W}^{(1)}_\sigma \xrightarrow{\sigma\mapsto\omega+\mathrm{Diff}} Z^{1,4}(s \mid \mathrm{d}) \longrightarrow H^{1,4}(s \mid \mathrm{d}). \tag{24}$$

The nonzero coordinates in Eq. (22) survive the last quotient. This is what promotes the effective-action Weyl anomaly to an obstruction to treating Weyl symmetry as an exact quantum gauge redundancy in the strict fixed-field-content theory.

Result card D: strict and compensated local quantum theories

Strict theory. Fixed-field-content Diff $\ltimes$Weyl pure C^2 gravity.

Domain. Complete declared gauge-fixed local BV quotient on a regular Bach-locus chart, with Euclidean one-loop coefficient input.

Statement. The regulated breaking has $c_{\rm W} = 199/30$, $a_{\rm W} = 87/20$, and vanishing parity-odd and chosen type-D coordinates. Its essential ghost-number-one, form-degree-four Weyl components are nonzero. The strict local Euclidean one-loop QME is obstructed.

Extended theory. In the formal τ-adic Wess–Zumino BV algebra with $s\tau = \omega$, the same cocycle is exact and a local counterterm restores the Euclidean QME at one loop.

Not claimed. The extension changes the BV theory. It gives no unconditional all-loop theorem, regulator/QAP for an arbitrary changed action, Lorentzian QME, Hadamard state, positive physical Hilbert space, particles, scattering, or unitarity.

The compensator statement is an exact BV realization of the familiar dilaton/Wess–Zumino mechanism, whose broader structure is known [33]. Its value is not a claim that a compensator is a free repair. Indeed, the passive compensator leaves a gauge-invariant dressed trace in the classical causal complex. Repairing that obstruction led to the changed counterflow experiment in Section 8.

Separately, the programme constructs candidate Hadamard two-point functions on a reduced free Einstein/additional/longitudinal mode space. The candidate pairing has signs $(+,-,-)$ on that selected space. This is useful evidence that an indefinite direction survives that reduction. It is not a BRST-compatible Hadamard covariance for the full metric BV complex and not a one-particle Hilbert space. No positive full-BV state has been constructed.

The immediate quantum literature includes conformal-gauge fixing and its additional ghosts, local BRST cohomology, and several alternative quantization prescriptions[31, 28]. Phase 1 neither reproduces all of that literature nor supersedes it. It contributes a content-addressed strict/extended local-BV calculation and records the precise point at which Lorentzian state construction remains absent.

8 What the counterflow candidate taught us

The strict anomaly and passive-compensator causal obstruction motivated a small changed-theory ladder. Passive trace repairs and a one-phase active clock failed combined causal, sign, and charge tests. Two phases with an auxiliary diagonal $U(1)$ allow neutral counterflow:

$$\dot{\chi} = 0, \qquad \dot{\psi} = \Omega, \qquad Q_{\text{diag}} = 0, \qquad E_{\text{rel}} = \frac{1}{2}\mu^2\Omega^2 > 0. \tag{25}$$

The relative phase can move while the diagonal compact gauge charge vanishes. Because its stress tensor is built from the dressed metric, it is quadratically visible in the dressed-trace direction.

For self-containment, the stationary family can be stated directly. Let σ_i be left-invariant one-forms on S^3 with

$$\mathrm{d}\sigma_1 = -\sigma_2 \wedge \sigma_3, \quad \mathrm{d}\sigma_2 = -\sigma_3 \wedge \sigma_1, \quad \mathrm{d}\sigma_3 = -\sigma_1 \wedge \sigma_2, \quad \int_{S^3} \sigma_1 \wedge \sigma_2 \wedge \sigma_3 = 16\pi^2. \tag{26}$$

In the gauge $\tau = 0$, $A = 0$, and constant χ, take

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(\sigma_1^2 + \sigma_2^2) + c^2\sigma_3^2, \qquad \psi = \Omega t, \tag{27}$$

and define

$$q = \frac{c^2}{a^2}, \qquad x = a^{-2}, \qquad C = \mu^2\Omega^2. \tag{28}$$

The exact stationary equations for Eq. (8) have rank three and give

$$M_2 = \frac{2C(4q-1)}{3qx}, \qquad V_0 = -\frac{C(q^2-5q+1)}{3q}, \qquad \alpha_B = \frac{2C}{qx^2}. \tag{29}$$

The connected causal trace-healthy component is

$$x > 0, \qquad C > 0, \qquad \frac{13-3\sqrt{17}}{4} < q < \frac{1}{4}. \tag{30}$$

The selected causal point fixes $\mu^2 = 1$ and

$$q = \frac{9}{40}, \quad x = 1, \quad C = \frac{9}{16}, \quad \Omega = \frac{3}{4}, \quad \alpha_B = 5, \quad M_2 = -\frac{1}{6}, \quad V_0 = \frac{119}{1920}. \tag{31}$$

The free BV object is the tangent complex of the displayed action and gauge transformations. Write its ghost and field bundles as

$$\mathcal{F}^{-1} = TM \oplus \mathbf{1}_{\mathrm{W}} \oplus \mathbf{1}_{U(1)}, \qquad \mathcal{F}^0 = S^2T^*M \oplus \mathbf{1}_\tau \oplus \mathbf{1}_\psi \oplus \mathbf{1}_\chi \oplus T^*M_A. \tag{32}$$

The minimal linear complex is then

$$\Gamma(\mathcal{F}^{-1}) \xrightarrow{q_{-1}} \Gamma(\mathcal{F}^0) \xrightarrow{q_0} \Gamma(\mathcal{E}) \xrightarrow{q_1} \Gamma((\mathcal{F}^{-1})^* \otimes \Lambda^4T^*M), \tag{33}$$

where q_{-1} is the linearized Diff–Weyl–$U(1)$ action, q_0 is the Euler–Lagrange Hessian, q_1 contains the Noether identities, and $\mathcal{E}$ is the dual equation bundle to the field bundle. The differential orders are also fixed by the action: q_{-1} is first order in the diffeomorphism and diagonal-$U(1)$ parameters and zeroth order in the Weyl parameter; q_0 is fourth order in the metric C^2 block, second order in the dressed Einstein and phase blocks, and algebraic in the auxiliary connection before elimination; q_1 has the corresponding formal-adjoint Noether orders. The cotangent BV rows and standard antighost–multiplier doublets complete this to a cyclic complex. The 70-row certificate is a harmonic-component implementation of Eq. (33); the invariant definition is the action and this graded complex.

Result card E: causal construction and physical nonselection

Theory. Equation (8) with the gauge transformations (9).

Background. The selected positive biaxial Berger solution and its connected trace-healthy same-field stationary retuning family.

Domain. The free cyclic BV complex (33); at the selected point its diagonal-$U(1)$ summand is contractible, and the physical spectral reduction uses complete $SU(2)_L \times U(1)_R$ isotypical blocks.

Gauge and charge. Diagonal gauge charge, relative-phase charge Q_{rel}, time translation D, and the background-preserving generator $K = D - \Omega R_{\mathrm{rel}}$ are kept distinct.

Causal statement. Exact retarded and advanced homotopies, nilpotency, cyclicity, and support identities hold at the selected point, and the original dressed-trace cohomology class is removed.

Physical statement. The complete both-k, $j = \frac{1}{2}$ reduced block contains a Hamiltonian–Hopf quartet. The obstructing factor is

$$(40z^4 + 773z^2 + 3748)^2, \qquad 773^2 - 4(40)(3748) = -2151 < 0.$$

With perturbations proportional to e^{zt}, the substitution $y = z^2$ gives two nonreal conjugate y-roots, hence four z-roots $\{\pm(\alpha+i\beta), \pm(\alpha-i\beta)\}$ with $\alpha\beta \neq 0$; two grow exponentially. The quartet persists throughout the connected trace-healthy same-field stationary family. Therefore no member of that family is selected as a robust linearly stable candidate.

Not claimed. This is not an all-isotype spectrum, a familywide Green-homotopy theorem, a full-PDE stability theorem, or a no-go over other field contents and gauge architectures.

The rejection criterion was fixed before the computation: a successor candidate had to be linearly stable in every physical sector. One exact unstable physical isotypical block is sufficient to reject it. It is not necessary to finish every higher-j block to establish this nonselection. Conversely, the finite block does not by itself describe the nonlinear evolution of arbitrary PDE data.

The persistence statement is exact rather than a numerical continuation from Eq. (31). Restoring $w = z^2/x$, the family characteristic divisor contains the quadratic factor

$$F_2(q, w) = 16q^2w^2 + 24q(3 - 2q^2)w + 32q^3 - 108q^2 + 81, \tag{34}$$

with discriminant

$$\operatorname{disc}_w F_2 = 256q^5(9q - 8) < 0 \qquad \text{on (30)}. \tag{35}$$

Thus the two w-roots remain a nonreal conjugate pair everywhere on the connected component; they cannot cross onto the negative real axis or pass through a multiple root. Since $x > 0$, their square roots remain a Hamiltonian–Hopf quartet. This no-crossing lemma is the familywide argument.

The quartet is a spectral Hamiltonian instability associated with a Krein-sign collision in the standard Hamiltonian–Krein sense [7], not merely a coordinate drift. The global clock block has a different status. On the unrestricted phase space,

$$\Omega_{\mathrm{rel}} = \mathrm{d}Q_{\mathrm{rel}} \wedge \mathrm{d}\psi, \qquad H_{\mathrm{rel}} = \frac{Q_{\mathrm{rel}}^2}{2I} + E_{\mathrm{geometry}}, \tag{36}$$

so its secular dephasing is tangent to an action–angle family and passes an orbital, frequency-modulated interpretation. It does not cure the $j = \frac{1}{2}$ quartet.

The charge analysis gives an independent theorem. For phase clocks with linear charge constraint $CQ = q_0$ and clock velocity vector v,

$$D \text{ is null after reduction} \quad \Longleftrightarrow \quad v \in \operatorname{im} C^T, \tag{37}$$

while physical clocks are classes of v modulo $\operatorname{im} C^T$. In the selected model, fixing and quotienting Q_{rel} makes D null but contracts the relative-clock Darboux pair. Keeping the clock leaves $H_D = \Omega Q_{\mathrm{rel}}$ and raw D charged; $K = D - \Omega R_{\mathrm{rel}}$ preserves the background. No single declared charge fibre supplies both the proposed physical clock and a gauge time translation.

The counterflow result is therefore not a failed piece of engineering. It separates three notions that are often conflated:

$$\begin{aligned} \text{causal completeness} &\quad \not\Rightarrow \quad \text{positive reduced dynamics}, \\ \text{positive reduced dynamics} &\quad \not\Rightarrow \quad \text{acceptable clock/charge interpretation}. \end{aligned} \tag{38}$$

The causal theorem remains valid even though the candidate is rejected by a later physical gate.

9 Integrated Phase-1 verdict

The main findings can now be placed on the four levels of Eq. (2).

Level	Established in a declared scope	Still absent
Local linear systems	Complete causal BV complexes on selected compact backgrounds; exact additional compact-product and Schwarzschild curvature solutions	A single causal theory covering every laboratory and boundary class
Reduced classical directions	Additional axial/polar compact-product directions are nonradical; counterflow causal parent and exact charge classification; radial-integrability Schwarzschild fixture	Universal asymptotic phase space, positive energy theorem, and acceptable counterflow spectrum
Nonlinear continuation	Five-charge necessary-and-sufficient finite exponential-polynomial cone; bounded resonance ledger and classified subcones	Full bounded finite-support zero locus, retarded nonlinear existence, all-order integration, and stability
Quantum states/particles	Strict local one-loop BV obstruction; formal changed-theory Wess–Zumino resolution; reduced indefinite two-point functions	Lorentzian QME, full-BV Hadamard state, positive physical Hilbert space, particles, scattering, optical theorem, and unitarity

This supports the following central claim:

> **A pole-level diagnosis is too coarse, but the more complete reduction does not generically rescue the theory.** Some additional classical directions survive gauge and presymplectic reduction. Nonlinear moment maps, resonances, boundary norms, local anomalies, and reduced spectral tests then eliminate different candidates for different reasons. The first changed theory to achieve a complete causal BV parent is rejected throughout its connected trace-healthy same-field family by a physical $j = \frac{1}{2}$ Hamiltonian–Hopf instability.

This is a classification ending, not a viable-theory claim and not a universal no-go. A different field content, a different gauge architecture, or a different admissible state prescription could define a new Phase 2. Phase 1 does not select one.

10 Relation to prior work

The programme combines several established structures. Its claims should be read as explicit constructions and exhaustive calculations in specified models, not as inventions of the surrounding general theories.

Causal gauge complexes. Green-hyperbolic complexes, retarded/advanced Green homotopies, and their induced Poisson structures have a general Lorentzian theory [11]. Conformal deformation and Yang–Mills detour complexes place the Bach operator in a longer conformal-geometric lineage [12, 13]. The contribution here is the explicit action-derived conformal-gravity BV construction, its cyclic pairings and support identities, and transfer across the declared backgrounds.

Linearization stability. The relation between Killing symmetries, quadratic Taub constraints, and singular momentum-map zero sets is classical [14, 15, 16, 17]. The finite-harmonic theorem is an exhaustive Weyl–Maxwell realization: exactly five covectors exhaust the exponential-polynomial cokernel, exceptional and generalized modes are included, and bounded resonance conditions are separated from the formal cone.

Einstein selection and black holes. Boundary-condition selection of Einstein solutions in conformal gravity is well known in Euclidean and asymptotically (A)dS settings [18, 19, 20]. Prior Weyl-black-hole quasinormal analyses often isolate a second-order conformally transported sector rather than the full fourth-order Bach solution space [21]. The static spherical Bach-flat lineage begins with Riegert and Mannheim–Kazanas[22, 23]. Four-dimensional conformal gravity already has consistent AdS boundary conditions, response functions, and finite canonical charges[24, 25]; our open problem is specifically the asymptotically flat/null-infinity Bach phase space. Our fixture-level radial-integrability result addresses the full additional reconstruction in a one-ended Lorentzian Schwarzschild problem, but does not yet provide a general asymptotically flat Bach phase space.

The magnetic product background belongs to the exceptional direct-product Einstein–Maxwell family introduced by Plebański and Hacyan [26]. The recent Weyl-gravity classification of separated $2+2$ direct products with electromagnetic and Yang–Mills sources proves generalized Birkhoff statements and organizes their Weyl-equivalence classes [27]. The present contribution is different: it fixes one compact magnetic bundle and computes its perturbative exact sequence, presymplectic form, nonlinear Taub cone, and resonances.

Quantum conformal gravity. One-loop divergences, conformal anomalies, BRST cohomology, and conformal gauge fixing have extensive prior literatures [29, 28, 31, 32]. The phase-one quantum claim is narrower: an exact local-BV quotient and coefficient ledger for the declared strict theory, followed by a formal τ-adic Wess–Zumino trivialization in a changed theory. It does not settle the competing PT-symmetric, Dirac–Pauli, fakeon, or conventional positive-state proposals.

Relational clocks. The construction of complete or partial observables from internal clocks is standard constrained-system territory[34]. The counterflow experiment adds a typed distinction between diagonal gauge charge, relative clock charge, time translation D, and the background-preserving combination K. No operational counterflow frequency ratio survived Phase 1, so the result is a charge/clock classification rather than an observational prediction.

11 Limitations and the next scientific boundary

The classification leaves several sharp, finite questions. The most important are not "finish quantum gravity." They are:

1. construct a complete asymptotically flat Bach phase space and determine which additional black-hole solutions have finite flux for general frequency and angular momentum;
2. determine the common zero locus of all bounded nonlinear obstruction functionals on the full declared finite-support space;
3. decide whether the retained mixed interaction defines a nonzero class in the complete bounded cyclic deformation complex. The interaction found in the first calculation can be removed, through the second tested order in the input fields, by the enlarged class of allowed field redefinitions. Therefore the earlier formula is not yet an invariant physical interaction;

4. construct a BRST-compatible Hadamard covariance for a full Lorentzian BV complex, or prove a scoped obstruction;

5. if a new changed theory is proposed, require an open stationary locus with causal completeness, positive reduced dynamics, coherent charge interpretation, a preserved nonlinear sector, and at least one operational observable before calling it a candidate.

These questions define a possible Phase 2. They are not conditions for closing Phase 1. The stopping rule was outcome-independent: Phase 1 ended when the programme could give an exact viability classification of the first causally complete changed-theory candidate and reconcile it with the compact linear, nonlinear, black-hole, observer, and quantum results. That record is now frozen.

12 Computational accountability

The project treats computer algebra as an experimental instrument whose outputs require independent checks. Material results are represented by a producer, a machine-readable certificate, a verifier that does not merely rerun the producer, adversarial mutation tests where practical, a tiered test receipt, and a human-readable scope report. Exact algebraic ranks, Smith forms, cohomology witnesses, and characteristic factors use rational or algebraic arithmetic; numerical evidence is typed separately.

The authoritative Phase-1 record is

`PURE_WEYL_PROGRAMME_PHASE1_CLASSIFICATION_ENDING_V1`,

stored in

`reports/phase1-closure-claims-ledger-2026-07-22.json`.

The public repository root is

https://github.com/area9innovation/bp2transformer/tree/master/physics/symplectic-reconstruction,

with this manuscript and its machine-readable claim map under https://github.com/area9innovation/bp2transformer/tree/master/physics/symplectic-reconstruction/paper. The JSON artifacts carry content hashes; no archival DOI is claimed at this stage. It binds each statement to a theory, background, phase space, correction or boundary class, lifecycle state, limitation, and source certificate. This paper has its own claim map and verifier; the live programme portal remains Paper 98. A claim that is not linked to evidence is left unpromoted rather than inferred from neighboring calculations.

Three aspects of this workflow matter scientifically. First, a failed promotion is retained: the invalid Berger round-sphere Hodge subspace, for example, was replaced by full $SU(2)_L \times U(1)_R$ isotypical blocks rather than silently repaired. Second, causal, spectral, algebraic, and quantum results carry different dependency labels. Third, stopping rules are fixed before the result is known. The counterflow candidate passed its causal gate and then failed a later physical gate; both facts remain part of the record.

13 Conclusion

The phase-one programme began with a common but underspecified question: whether the additional poles of pure Weyl gravity are physical ghosts. Its main conceptual result is that there is no single promotion from pole to particle. Local solutions, reduced classical directions, nonlinear tangent directions, and quantum states are different mathematical objects.

That distinction did not automatically rescue the theory. Additional compact-product directions survive local reduction, yet some fail bounded nonlinear continuation. Horizon regularity admits additional Schwarzschild curvature solutions, yet a specified radial-integrability fixture selects the Einstein sector. Strict pure Weyl gravity has a local Euclidean one-loop BV obstruction; a compensator removes it only in a changed theory. The first changed model with a complete causal BV parent is itself rejected by an exact family-persistent Hamiltonian–Hopf instability.

The most defensible conclusion is therefore neither "the propagator already settled the physics" nor "constraints removed every ghost." It is:

Pole-level signs are an initial diagnostic. Physical viability is a sequence of reductions and existence tests. In the laboratories completed in Phase 1, different additional sectors survive different early tests, but no candidate reaches a positive full quantum state, and the first causally complete changed theory fails a robust linear physical-stability test.	(39)

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